

MIEL GABRIEL LAGDAN

Applied Physics Undergraduate | Theoretical Physics Interests

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PROFESSIONAL SUMMARY

An applied physics undergraduate with a strong foundation in mathematical and theoretical physics, complemented by experience in undergraduate-level instruction and research. Knowledgeable in computational tools such as MATLAB, R, and Python, with research interests in theoretical and mathematical physics, particularly quantum theory and particle physics.

EXPERIENCE

VOLUNTEER TUTOR

University of the Philippines Manila
Ermita, Manila, Philippines

October 2023 – Present
PART-TIME

- Provides support to 30+ undergraduate students seeking reinforcement in learning topics involving algebra, calculus, real analysis, theoretical mechanics, electromagnetism, and quantum mechanics.
- Produces numerous helpful resources, including sample examinations, that helped many students in honing their skills in solving problems involving analysis and physics.
- Conducts one-to-one or many-to-one tutorial sessions to the students at the University of the Philippines Manila free of charge and helps other tutors with assistance in procuring resources for their students.

TUTOR

Great Minds Tutorial and Review Center
Quezon City, Philippines

August 2025 – Present
PART-TIME

- Provides supplementary help to 10+ elementary and high school students in topics related to mathematics and science (specifically physics) and recommends studying regimens tailored to each student.
- Creates and procures helpful materials in pedagogy useful to the student and the tutoring community.
- Assists students in preparation for upcoming academic competitions, nationally and internationally.

RESEARCH EXPERIENCE

Energies of two-flavor neutrino oscillations in uniform mass distribution using Dirac equation

2025

Proceedings of the Samahang Pisika ng Pilipinas

Miel Gabriel Lagdan, Jourdyn Roland N. Garcia, Herbert B. Domingo

- Performed theoretical analysis of neutrino oscillations of particles of uniform masses using coupled Dirac equations.
- Used Elisabetta Sassaroli’s model (1999) of neutrino oscillations and extended it by applying small mass increments.
- Provided a framework for uniform mass distribution models, with recommendations for extensions for varying mass.

Assessment of the Effectiveness of “Thermika,” a Serious Game, in Teaching Thermochemistry

2024

Philippine Federation of Chemistry Societies, Inc.

Miel Gabriel Lagdan, Pauline Khyla D. Gabatin, Niño Sherwin Lumancas, Roma Balagtas-Arguelles, Toshi Akira Capintog

- Development and statistical analysis of a serious game’s effectiveness to 84 high school students studying thermochemistry.
- Results demonstrate statistically significant and effective application of intervention with statistical evidence.
- An online, playable version was uploaded to itch.io, where students are free to access the game to assist in learning.

EDUCATION

Bachelor of Science in Applied Physics (Health Physics Concentration)

University of the Philippines Manila

Expected Graduation: 2027
August 2023 – Present

- With relevant coursework in radiation physics, statistical methods, magnetic resonance imaging, and computational physics.
- Joined a research laboratory in the University of the Philippines Manila called “Laboratory for Applied Mathematical Physics.”

Secondary Education

Caloocan City Science High School

September 2021 – July 2023

SKILLS

Technical Skills:

Python (numerical analysis)
MATLAB & R (statistical analysis, simulations)
LaTeX (technical writing)
Arduino (C++) (hardware interfacing)

Analytical Skills:

Mathematical modeling of physical systems
Differential equation analysis (ODEs & PDEs)
Approximation techniques
Statistical techniques

Other Skills:

Communication
Teamwork
Adaptability
Leadership